

STANDARD OPERATING PROCEDURE

SOP-01: Pulmonary Function Testing Execution

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Department: Clinical Operations
Approved By: Robert E. Beaty, Founder

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Modality: Mobile Diagnostics

1.0 Purpose & Scope

1.1 Purpose: To establish uniform, evidence-based clinical guidelines for the execution of comprehensive Pulmonary Function Testing (PFT) utilizing the PulmOne MiniBox+ diagnostic suite. This procedure ensures all diagnostic data captured in the field meets the rigorous acceptability and repeatability criteria established by the American Thoracic Society (ATS) and the European Respiratory Society (ERS).

1.2 Scope: This protocol applies to all mobile clinical testing performed by credentialed Direct-Care Pulmonary Diagnostics personnel within partner primary care clinics, Rural Health Clinics (RHCs), and industrial screening sites.

2.0 Clinical Contraindications Screening

Prior to testing, the supervising Registered Respiratory Therapist (RRT) must verify the patient's medical history via the partner clinic EMR or direct patient interview to rule out absolute and relative contraindications. Maximal forced exhalation maneuvers and lung volume measurements place significant stress on the thoracic, abdominal, and ocular cavities.

2.1 Absolute & Relative Contraindications (Delay Testing If Present):

- **Cardiovascular:** Myocardial infarction within the last 1 month; unstable angina; recent history of syncope related to forced exhalation/coughing; known thoracic, abdominal, or cerebral aneurysm.
- **Surgical:** Recent thoracic, abdominal, or eye surgery (e.g., cataract removal) within the past 1 to 3 months where increased intra-thoracic or intra-ocular pressure presents a rupture risk.
- **Respiratory:** Active hemoptysis of unknown origin; recent pneumothorax; active respiratory infection (e.g., active Tuberculosis, symptomatic COVID-19).

3.0 Patient Preparation & Posture

1. **Verification:** Confirm patient identity using two identifiers (Name and Date of Birth).

2. **Biometrics:** Accurately measure and input the patient's current height, weight, age, and biological sex assigned at birth into the MiniBox+ software to ensure correct predicted-value reference sets (e.g., GLI-2012).
3. **Posture:** Patient must be seated in a stable, upright chair with arms, feet resting flat on the floor. Wheelchair-bound patients should lock brakes. Neck must be slightly elevated to ensure an unobstructed airway.
4. **Infection Control Setup:** Attach a new, single-use 0.025-micron viral/bacterial barrier filter to the flow sensor. Apply medical-grade nose clips to the patient to ensure exclusive mouth breathing.

4.0 Procedure Execution (MiniBox+)

4.1 Gasless Lung Volumes (TLC, FRC, RV)

- Instruct the patient to seal their lips tightly around the filter mouthpiece.
- Initiate normal, relaxed tidal breathing (Functional Residual Capacity baseline).
- At end-expiration, the software will briefly actuate the shutter valve. Instruct the patient to perform gentle panting maneuvers against the closed occlusion.
- Following the shutter release, instruct the patient to take a maximal deep inspiration to Total Lung Capacity (TLC), followed by a complete exhalation to Residual Volume (RV).

4.2 Full Spirometry (Pre/Post Bronchodilator)

- Instruct patient to inhale maximally and completely until the lungs are entirely full.
- Command a forceful, explosive exhalation ("Blast!") and encourage the patient to keep pushing air out for a minimum of 6 seconds (or until a definitive plateau is reached on the volume-time curve).
- Prompt a rapid, forceful maximal inspiration back to baseline to complete the flow-volume loop.
- *For Post-Bronchodilator Testing:* Administer the prescribed short-acting bronchodilator (e.g., Albuterol). Wait 10 to 15 minutes. Execute formal Inhaler Technique Education (CPT 94664) during this wait period. Repeat spirometry.

4.3 Single-Breath DLCO (Diffusing Capacity)

- Instruct patient to breathe normally, then exhale completely to RV.
- Command a rapid, deep inspiration of the calibration gas mixture (containing tracer gases) to TLC.
- Instruct the patient to hold their breath cleanly (no Valsalva or Müller maneuvers) for approximately 10 seconds.

- Command a smooth, continuous exhalation to allow the infrared sensor to sample the alveolar gas.

5.0 Quality Assurance & ATS/ERS Acceptability

The Supervisory RRT must visually inspect flow-volume loops and system quality grades before concluding the exam.

- **Spirometry Repeatability:** Obtain a minimum of three (3) acceptable spirograms. The two largest FVC and FEV1 values must be within 150 mL of each other.
- **DLCO Repeatability:** Obtain at least two (2) acceptable DLCO maneuvers. The values must be within 2 mL/min/mmHg (or 10%) of each other. Allow at least 4 minutes between DLCO maneuvers for gas washout.
- **Data Sync:** Upon completion, the RRT will finalize the testing packet and initiate secure cellular sync to the cloud portal for pulmonologist tele-overread.

END OF STANDARD OPERATING PROCEDURE SOP-01